5.3 TYPES OF SPECTRUM- EMISSION AND ABSORPTION SPECTRUM- FRAUNHOFER LINES

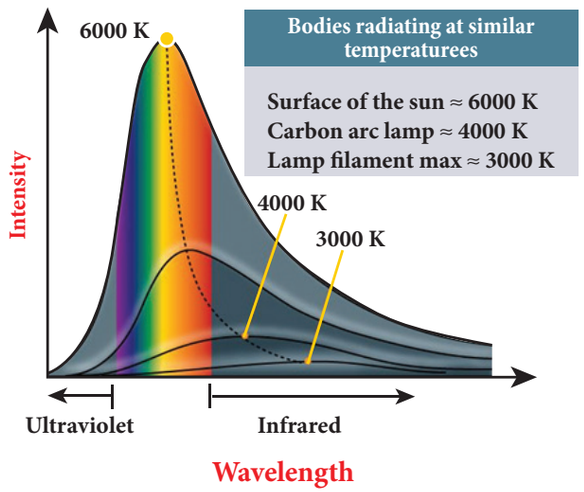


Figure 5.11 Black body radiation spectrum – variation with temperature

When an object burns, it emits colours. That is, it emits electromagnetic radiation which depends on temperature. If the object becomes hot then it glows in red colour. If the temperature of the object is further increased then it glows in reddish-orange colour and becomes white when it is hottest. The spectrum in Figure 5.11 usually called as black body spectrum (refer plus one volume two Unit 8). It is a continuous frequency (or wavelength) curve and depends on the body's temperature.

Suppose we allow a beam of white light to pass through the prism as shown in Figure 5.12, it is split into its seven constituent colours which can be viewed on the screen as continuous spectrum. This phenomenon is known as dispersion of light and the definite



Figure 5.12 White light passed through prism – dispersion

pattern of colours obtained on the screen after dispersion is called as spectrum. The plural for spectrum is spectra. The spectra can be broadly classified into two catagories:

(a)Emission spectra

When the spectrum of self luminous source is taken, we get emission spectrum. Each source has its own characteristic emission spectrum. The emission spectrum can be divided into three types:

(i) Continuous emission spectra (or continuous spectra)



Figure 5.13 continuous emission spectra

If the light from incandescent lamp (filament bulb) is allowed to pass through prism (simplest spectroscope), it splits into seven colours. Thus, it consists of wavelengths containing all the visible colours ranging from violet to red (Figure 5.13). Examples: spectrum obtained from carbon arc, incandescent solids, liquids gives continuous spectra.

(ii) Line emission spectrum (or line spectrum):

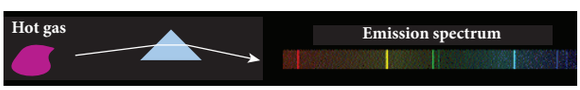


Figure 5.14 line emission spectra

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